Figure 3. Accuracy of frequency estimates

Frequency = average events among people with at least one event

Poisson | Negative binomial (NB2) | Hurdle NB2

Mean estimation error (RMSE, events per affected person)

No stratum interaction effects — Occurrence 0.5% | Frequency 1.2
Common fits: 283 | Common fits: 447
100 affected N = 20,000 | 500 affected N = 100,000

No stratum interaction effects — Occurrence 2% | Frequency 1.2
Common fits: 295 | Common fits: 435
100 affected N = 5,000 | 500 affected N = 25,000

No stratum interaction effects — Occurrence 0.5% | Frequency 3
Common fits: 499 | Common fits: 500
100 affected N = 20,000 | 500 affected N = 100,000

No stratum interaction effects — Occurrence 2% | Frequency 3
Common fits: 499 | Common fits: 500
100 affected N = 5,000 | 500 affected N = 25,000

Interaction effects: same direction — Occurrence 0.5% | Frequency 1.2
Common fits: 311 | Common fits: 467
100 affected N = 20,000 | 500 affected N = 100,000

Interaction effects: same direction — Occurrence 2% | Frequency 1.2
Common fits: 318 | Common fits: 451
100 affected N = 5,000 | 500 affected N = 25,000

Interaction effects: same direction — Occurrence 0.5% | Frequency 3
Common fits: 499 | Common fits: 500
100 affected N = 20,000 | 500 affected N = 100,000

Interaction effects: same direction — Occurrence 2% | Frequency 3
Common fits: 500 | Common fits: 500
100 affected N = 5,000 | 500 affected N = 25,000

Interaction effects: opposite directions — Occurrence 0.5% | Frequency 1.2
Common fits: 321 | Common fits: 453
100 affected N = 20,000 | 500 affected N = 100,000

Interaction effects: opposite directions — Occurrence 2% | Frequency 1.2
Common fits: 347 | Common fits: 445
100 affected N = 5,000 | 500 affected N = 25,000

Interaction effects: opposite directions — Occurrence 0.5% | Frequency 3
Common fits: 500 | Common fits: 500
100 affected N = 20,000 | 500 affected N = 100,000

Interaction effects: opposite directions — Occurrence 2% | Frequency 3
Common fits: 499 | Common fits: 500
100 affected N = 5,000 | 500 affected N = 25,000

Expected number affected; N is the total sample size

Equal weight per stratum. All three models use the same acceptable datasets within each setting.

Bars: approximate 95% Monte Carlo intervals for mean RMSE, not uncertainty in patient outcomes.